



Scenario (New Product Development)

Overview:

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Background:

This is a common scenario faced by virtually every organization. We'll use a small medical device manufacturer for our discussion since it crosses many sectors and allows us to explore how EA can be used to create a foundation for guiding and enabling growth. Growth typically comes in a few approaches (new product development = NPD; new partnerships, or product/market acquisition). The goal is to define and implement a framework for driving sustainable growth. This scenario in brief will cover:

- a. Opportunity Assessment to outline the areas of focus for both the business and the enterprise architecture areas.
- b. Current State Assessment of the business goals, opportunities and challenges
- c. Future State Design outlining a framework for achieving the business goals by guiding the right business change atop a foundation of structures involving people, processes, and technologies.
- d. Impact Assessment reviews the changes that have been achieved for the business and enterprise architecture areas of focus.

Opportunity Assessment:

Driving growth through New Product Development (NPD) is one of the most challenging approaches in business (particularly in heavily regulated environments like healthcare). More companies choose the other options instead of laying the proper foundation for delivering sustainable value typically associated with NPD. This scenario requires a broad range of skills and experience from an EA to successfully guide an organization along the journey. Below is a short list of the Business Domains and stakeholders that would need to be involved in defining the requirements and approving the solution.

Business Domain Analysis:

Business Domain	Key Stakeholders
Product Management	Chief Marketing Officer
Information Technology	Chief Information Officer
R&D	Chief Scientific Officer
Program Management	PMO
Operations	Plant Manager(s)
Customers & Shareholders	CEO

The following list of EA Domains that are involved in defining the framework.

EA Domain Analysis:

- Business Architecture
- Information Architecture
- Application / Solution Architecture



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- Implementation Architecture
- Operational Architecture

Current State Assessment:

Background:

The organization (SMB Device Mfg) has cut back its expenses over the last few years to strengthen their cash reserves and position itself for growth. After a brief period of contraction and preservation, the organization is preparing for growth and expansion. The senior leadership team has decided to pursue an NPD strategy by investing in new product lines and technology. However, their past attempts at developing new products have resulted in only 13% of these products being successful. A top tier management consulting firm was brought in to advise the organization on NPD, and overall organizational direction. The price tag for the changes was prohibitively expensive, so the senior leadership team asked a small (but trusted) firm to perform an EA assessment and recommend a future state direction within a capped budget over 3-yr period. Below are some of the key findings during the assessment:

Product Management:

- Marketing, Product Mgmt, and R&D operate in silos
- Customer Focus is splintered and lacks clarity of view
- Lacks of a model-driven view for conceptualizing & enhancing products and features
- Low % of product launch successes (~15%)

Information Technology:

- Customer (internal & external) needs are captured in documents instead of in a repository
- Product requirements are not easily derived (traceable) from External Customer needs
- Operational requirements are not easily derived from Internal Customer needs
- Product features are not aligned with prioritized view of customer needs

R&D / Product Dev:

- Engineering Specifications are not aligned with Customer needs & Product requirements
- Product concepts often need to be re-worked when reviewed by customers
- Alpha & Beta testing cycles are longer than competitors for similar products
- Product launches go through lengthy re-factoring until the product is “perfected”
- Product software is developed using an “agile” methodology to accommodate frequent changes (late in the development cycle)

Program Management:

- Product costs frequently exceed budgets
- Product launches often miss the target dates (or key features are sacrificed)
- Release management often misses the market windows (time-to-market 4-7 months longer than competition)



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Operations:

- Plant Managers are incentivized to manufacture large volume orders which conflict with the trend towards mass customization for customers (customer base has become fragmented as hospitals are divesting many of their procedures to alternate care centers – whose needs are very different)
- The device architecture was designed with a tightly integrated set of components, which makes changes (for customers) expensive and time consuming.

Future State Design:

The enterprise architect reviewed the above findings with the senior leadership team to obtain validation and support for Gap Analysis (see below). The Gap Analysis was subsequently reviewed as part of a foundational walk-through to educate the key stakeholders and obtain their buy-in for preparing a blueprint and roadmap of investments to address the business concerns (stated above). This is where an EA's experience across a multitude of disciplines and domains creates separation from other consultants and professionals.

Gap Analysis (EA Domains):

There is a greatly simplified overview of the actual analysis as this is a scenario outline.

- No formal Business Architecture exists to support a data driven approach for modelling customer needs and transitioning them into engineering specifications
- Requirements, Data Warehousing, and Pattern-based extraction were not in place for rationalizing the Information Architecture to support an efficient product development process
- Lack of interoperability and limited flexibility indicated an underperforming Application / Solution Architecture
- There is no formal requirements engineering process in place, which results in project slippage and missed commitments. In addition, there is no agreed upon process across Marketing, R&D, and Product Mgmt for prioritizing projects let alone product features. The Implementation Architecture lacks discipline and formal structure to support the existing (let alone future) change management needs.
- Product design does not support the rapid shift from high volume manufacturing to mass customization of medical devices to meet customer needs. The Operational Architecture lacked a formal platform from which to drive flexibility and extensibility to support rapid change (in design) and product delivery.

Once the Gap Analysis was completed, the EA conducted a series of focused ideation sessions with key stakeholders and selected domain experts to identify a list of ideas, which would form the basis for solutions to address each of the business challenges (stated above).

Blueprinting & Justification:

Once the Gap Analysis is reviewed, the EA facilitated a series of discussions with domain architects (if available w/in the organization) to outline a series of projects to address the architecture deficiencies. These projects included various types of transition points covering people, processes, and technology

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across the business & EA domains. The list of projects was then reviewed with the key stakeholders (and/or appointed decision makers) to explain how these changes (projects) would address the underlying gaps and enable the business to address the business challenges. This was a key step in building understanding and support for addressing the immediate EA challenges (within the scope of the roadmap), but also to ensure there was ongoing support for sustaining the EA (once the roadmap was completed). Once the business was able understand the cause and effect relationships between EA structures and achieving desired business outcomes, business case development for the projects was a straight forward exercise.

Prioritization & Roadmapping

- The full list of projects were then prioritized and laid out in a multi-year roadmap
- Funding models were put in place to support the progressive list of investments
- Measurement models were established to align business & EA concerns
- Existing structures and processes were enhanced to ensure the investments in EA and business capabilities were properly planned, funded, and governed (during and after implementation).

Impact Assessment:

The following is a brief review of the results that were achieved & lessons learned:

Product Management:

- A formal discipline for customer focus was implemented for one product line (others to follow as the process matures)
- Products are now conceptualized within a model-driven framework that aligns concerns for sales, marketing, finance, engineering, and product support.
- Product planning and project portfolio investments are now aligned using a model that aligns priorities across marketing, product management, and R&D.
- Product launch success rate has increased to 60% (still rising as the discipline around customer focus matures and expands to other product lines)

Information Technology:

- A requirements engineering discipline was implemented to capture and warehouse customer needs using a defined semantic and data model.
- Internal & External Customer needs are integrated into the data model
- Requirements and Feature Decomposition are modelled and aligned in the requirements repository
- Data Management tools were extended to provide stewardship over the data and identify patterns for driving new opportunities to expedite product development
- Metrics are now modelled across the change management processes and systems to ensure quality, efficiency, and a balanced alignment between customer needs, compliance, and business concerns (tactical) is maintained.



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R&D / Product Dev:

- Engineering Specifications are now derived from the requirements repository and traceability is assured from specification through product testing.
- Concept to market cycle time has seen a 40% reduction in the initial set of projects for NPD.
- Alpha & Beta testing cycles has seen a 60% reduction in the number of defects.
- Initial products that were launched with the new framework have been very smooth (no recalls thus far)

Program Management:

- Project commitments are not being met for those projects transitioned into the new change management framework
- Marketing has been able to meet their initial target dates for planned releases (for those few products launched within the one product line). Limited rollout is slowly expanding as teams and processes are transitioned over.
- New governance processes are much more efficient and now rely on the operational metrics using the improved data and metrics models that span across the product development lifecycle through release management.

Operations:

- Plant Management has shifted incentives for line operations to support a customer-driven model versus the operationally efficient model, which was previously in place.
- Line operations were revamped and integrated into an e-batch system to support an on demand capability for order fulfilment. (Note: These processes are some of the biggest and most complex to change when part of an existing ERP).

Lessons Learned:

- Ensure Operations are included in up-front discussions as they can impact downstream effectiveness when planning and implementing product changes.
- Include plant engineering in product ideation going forward as their input can influence product design in positive ways that provides mutual benefit to operations and value chain customers.
- Ensure metrics are captured between the front-end and back-end as these transformations take hold over time. This will ensure the focus stays on the sustainable value instead of the short-term (investments in the roadmap and new processes).